

ACADEMIC DETAILS

Degree	Specialization	Institute	Year	CPI/%
M.S.	Program in Integrated Circuit Design	National Cheng Kung University (NCKU), Tainan, Taiwan (Double Degree)	2026–Present	–
M.Tech.	Integrated Circuit Design and Technology	IIT Gandhinagar	2025–Present	7.6
B.Tech.	Electronics Engineering	Birla Vishvakarma Mahavidyalaya	2021–2025	9.16
Class XII	Science Stream (GSEB)	Mother of Hope (Ashadham) School	2020	59.38
Class X	General Stream (GSEB)	Shri Vidhya Niketan	2018	75.33

INTERNSHIPS

- **ASIC Intern, eInfochips (An ARROW Company)**

[Jan '25 - Jun '25]

(Ahmedabad, India)

◦ Designed and verified digital IP blocks using Verilog and SystemVerilog targeting standard ASIC design flows.

◦ Implemented complete AMBA APB Protocol master–slave RTL architecture and state machine control logic.

◦ Built a modular SystemVerilog verification environment (Generator, Driver, Monitor, Scoreboard) for synchronous FIFO.

◦ Achieved 94% functional and code coverage by verifying edge cases, boundary conditions, and cross-coverage metrics for synchronous FIFO.

PROJECTS

- **Design and Implementation of Oscillatory Neural Network on FPGA**

[Dec '25 - Jul '26]

(Prof. Sandip Laskare, IIT Gandhinagar) | [Project Link](#)

◦ Developed an Oscillatory Neural Network (ONN) using coupled oscillators for low-power, high-speed pattern recognition.

◦ Implemented architecture on Zybo Z7-20 FPGA (Zynq-7020 PS/PL) using parallel hardware computation for high throughput.

◦ Designed a hardware-accelerated pipeline to classify matrix-based patterns in real time with minimal latency.

◦ Built a custom bootable image and driver stack with PetaLinux for efficient hardware–software co-design.
- **5-Stage Pipelined RISC-V Processor**

[Jan '26 - May '26]

(Prof. Joycecie Mekie, IIT Gandhinagar) | [Project Link](#)

◦ Designed a 5-stage pipelined RISC-V processor in SystemVerilog (Fetch, Decode, Execute, Memory, Writeback).

◦ Integrated forwarding units and hazard detection logic to resolve data, control, and structural hazards with minimal stalls.

◦ Built comprehensive SystemVerilog testbenches to verify instruction execution flow and pipeline timing accuracy.
- **AES-128-based Secure UART Communication System on FPGA**

[Sept. 24 - Dec. 24]

(Prof. Anita Bhatt, Birla Vishvakarma Mahavidyalaya) | [Project Link](#)

◦ Designed a hardware-accelerated secure UART system integrated with an AES-128 cryptographic engine.

◦ Developed Verilog RTL for AES cipher stages (Key Expansion, SubBytes, ShiftRows, MixColumns) and custom UART RX/TX transceivers for data reception and transmission.

◦ Validated dual-board Xilinx Artix-7 Basys 3 FPGA communication, demonstrating real-time end-to-end encryption and decryption.

TECHNICAL SKILLS

- **Hardware Description Languages:** Verilog, SystemVerilog

• **EDA Tools & Simulators:** Xilinx Vivado, Vitis, Cadence Virtuoso, Synopsys VCS, TCAD

• **Protocols & Architectures:** AMBA APB, AXI4 - Lite, UART, RISC-V Architecture

• **FPGA & Embedded Systems:** Zybo Z7-20 (Zynq SOC), Artix-7 Basys 3, PetaLinux

ACHIEVEMENTS

- **Winner – National Robotics Competition (Robofest 4.0):** Secured 1st place in the Rover category by developing an autonomous robotic control system.

EXTRA-CURRICULAR ACTIVITIES

- **Sports:** Active tennis player, focusing on endurance, quick decision-making, and teamwork.

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant – Digital System Design, IIT Gandhinagar**

[Aug '25 - Nov '25]

◦ Guided 50+ undergraduate students in Verilog HDL programming, FPGA synthesis, and hardware debugging.

◦ Facilitated laboratory sessions focusing on RTL design methodology, finite state machines (FSMs), and timing logic.